

# ZIHAO WANG

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## INTERESTS

My research centers on building open-ended generalist agents, including computer-use agents, embodied game agents, and MCP-based agents. My core interest lies in building and leveraging large pre-trained Foundation Models (LLMs, VLMs, VLA) to significantly enhance agent generalization capabilities. My research contributions fall into two main categories:

- **Agentic Foundation Models:** **UI-TARS-1.5** (An open-source multimodal GUI and Game agent built upon a powerful vision-language model), **OmniJARVIS** (Hierarchical VLA with latent action space), **JARVIS-VLA** (A vision-language-action model in open worlds).
- **Agentic Workflow:** **DEPS** (Planning Agent), **JARVIS-1** (Self-improving Agent with Multimodal Memory), **RAT** (Open Deep Research Agent), **ROCKET-1** (Embodied Agent with GUI action space), **ProAgent** (Collaborating LLM-based Game Agents).

## EDUCATION

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|---------------------------------------------------------------|-----------------------|
| <b>Peking University</b>                                      | <b>Beijing, China</b> |
| <a href="#">Ph.D.</a> Student in Artificial Intelligence (AI) | 2022.09 - Present     |
| - Advisor: Professor Yitao Liang                              |                       |
| <b>Beijing Institute of Technology</b>                        | <b>Beijing, China</b> |
| <a href="#">M.S.</a> in Control Science and Technology        | 2019 - 2022           |
| - Advisor: Professor Zhen Li                                  |                       |
| <b>Beijing Institute of Technology</b>                        | <b>Beijing, China</b> |
| <a href="#">B.S.</a> in Automation                            | 2015 - 2019           |

## PROFESSIONAL EXPERIENCE

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|---------------------------------------------------------|-------------------|
| Research Intern, Seed in Bytedance Inc, Beijing, CHINA  | 2024.12 - 2025.05 |
| Research Intern, BIGAI, Beijing, CHINA                  | 2022.06 - 2024.09 |
| Research Intern, AI Lab in Ailibaba Inc, Beijing, CHINA | 2021.06 - 2021.08 |

## PUBLICATIONS

The symbol \* indicates the same contribution.

### • Pre-Trained Agent Models

- [1] [Seed Tars Team](#). UI-TARS-2 Technical Report: Advancing GUI Agent with Multi-Turn Reinforcement Learning. [\[Showcases\]](#) [\[Paper\]](#)
- [2] [Seed Tars Team](#). UI-TARS-1.5: Pioneering Automated GUI Interaction with Native Agents. [\[Blog\]](#) [\[Huggingface Models\]](#)
- [3] [ByteDance Seed](#). Seed1.5-VL Technical Report. *arXiv preprint arXiv:2505.07062*, 2025
- [4] [Zihao Wang](#), ..., Anji Liu, Xiaojian Ma, Yitao Liang. OmniJARVIS: Unified Vision-Language-Action Tokenization Enables Open-World Instruction Following Agents. **NeurIPS 2024**. [\[Page\]](#)
- [5] [Muyao Li\\*](#), [Zihao Wang\\*](#), ..., Yitao Liang. JARVIS-VLA: Post-Training Large-Scale Vision Language Models to Play Visual Games with Keyboards and Mouse. **ACL 2025** [\[Page\]](#)

### • VLM/LLM-Based Agent: Planning, Memory and Reasoning

- [6] Zihao Wang, ..., Yaodong Yang, Xiaojian Ma, Yitao Liang. JARVIS-1: Open-world multi-task agents with memory-augmented multimodal language models. **T-PAMI, 2024**. [\[Page\]](#)
- [7] Zihao Wang, ..., Anji Liu, Xiaojian Ma, Yitao Liang. Describe, Explain, Plan and Select: Interactive Planning with Large Language Models Enables Open-World Multi-Task Agents. **NeurIPS, 2023**; also **Best Paper on ICML 2023 TEACH Workshop**. [\[Code\]](#)
- [8] Zihao Wang, ..., Yitao Liang. RAT: Retrieval Augmented Thoughts Elicit Context-Aware Reasoning in Long-Horizon Generation. *Open-world Agents Workshop in NeurIPS 2024*. [\[Page\]](#)
- [9] Ceyao Zhang, Kaijie Yang, Siyi Hu, Zihao Wang, ..., Yitao Liang, Yaodong Yang. ProAgent: Building Proactive Cooperative Agents with Large Language Models. **AAAI 2024 (Oral)**. [\[Page\]](#)

- **Action Tokenizers in Hierarchical Agentic Models**

- [10] Jingwen Deng\*, Zihao Wang\*, Shaofei Cai, Anji Liu, Yitao Liang. Open-World Skill Discovery from Unsegmented Demonstrations. **ICCV, 2025**.
- [11] Shaofei Cai, Zihao Wang, Kewei Lian, Zhancun Mu, Xiaojian Ma, Anji Liu, Yitao Liang. ROCKET-1: Master Open-World Interaction with Visual-Temporal Context Prompting. **CVPR 2025**. [\[Page\]](#)
- [12] Shaofei Cai, Bowei Zhang, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. GROOT-2: Weakly Supervised Multi-Modal Instruction Following Agents. **ICLR 2025**. [\[Paper\]](#)
- [13] Shaofei Cai, Bowei Zhang, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. GROOT: Learning to Follow Instructions by Watching Gameplay Videos. **ICLR 2024 (Spotlight)**. [\[Page\]](#)
- [14] Shaofei Cai, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. Open-world multi-task control through goal-aware representation learning and adaptive horizon prediction. **CVPR, 2023**. [\[Code\]](#)

- **Agent Evaluations**

- [15] Haowei Lin, Zihao Wang, Jianzhu Ma, Yitao Liang. MCU: A Task-centric Framework for Open-ended Agent Evaluation in Minecraft. **ICML, 2025 (Spotlight)**.
- [16] Haowei Lin, Xiangyu Wang, Ruilin Yan, Baizhou Huang, Haotian Ye, Jianhua Zhu, Zihao Wang, James Zou, Jianzhu Ma, Yitao Liang. Generative evaluation of complex reasoning in large language models. arXiv: 2504.02810.

- **Visual Geometry and SLAM**

- [17] Zihao Wang, Chunxu Wu, Yifei Yang, Zhen Li. Learning Transformation-Predictive Representations for Detection and Description of Local Features. **CVPR, 2023**.
- [18] Zihao Wang, Zhen Li, Xueyi Li, Wenjie Chen, Xiangdong Liu. Graph-Based Contrastive Learning for Description and Detection of Local Features. **IEEE Trans. Neural Netw. Learn. Syst. (TNNLS) 2022**.
- [19] Zihao Wang, Xueyi Li, Zhen Li Local Representation is NOT Enough: Soft Point-wise Transformer for Descriptor and Detector of Local Features. **IJCAI 2021**.

- **Others**

- [20] Haowei Lin, Baizhou Huang, Haotian Ye, Qinyu Chen, Zihao Wang, ..., James Zou, Yitao Liang. Selecting Large Language Model to Fine-tune via Rectified Scaling Law. **ICML 2024**.
- [21] Yuheng Cheng, Ceyao Zhang, Zhengwen Zhang, Xiangrui Meng, Sirui Hong, Wenhao Li, Zihao Wang, Zekai Wang, Feng Yin, Junhua Zhao, Xiuqiang He. Exploring large language model based intelligent agents: Definitions, methods, and prospects. *arXiv:2401.03428, 2024*.
- [22] Haotian Zhang, Junting Zhou, Haowei Lin, ..., Zihao Wang, Liangcai Gao, Yizhou Wang, Yitao Liang. CLoG: Benchmarking Continual Learning of Image Generation Models. *arXiv preprint arXiv:2406.04584, 2024*. [\[Code\]](#)
- [23] Guangyu Zhao, Kewei Lian, Haowei Lin, Haobo Fu, Qiang Fu, Shaofei Cai, Zihao Wang, Yitao Liang. Optimizing Latent Goal by Learning from Trajectory Preference. *arXiv preprint arXiv:2412.02125, 2024*.

## AWARDS

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- **Principal Scholarship** (Highest Scholarship for Phd Student in Peking University), CHINA, *2025*.
- **Ranking 2nd** on ECCV 2024 Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving Challenges, *2024*.
- **Best Paper** in ICML 2023 TEACH Workshop, *2023*.
- **National Scholarship**, CHINA, *2021*
- **Outstanding Graduate** (top 1%), Beijing, CHINA, *2019*
- **Special Prize** on Autonomy in International Micro-unmanned Aerial Vehicle Competition (IMAV), Melbourne, Australia, *2018*